1 Construction of the set of natural numbers

Let us build, using Peano's axioms, the set of **natural numbers** $\mathbb{N}$.

- **Reflexivity**: $a = a$.
- **Symmetry**: $a = b \iff b = a$.

Axiom 1. 1 is a natural number, i.e. $1 \in \mathbb{N}$.

Axiom 2. For every $x \in \mathbb{N}$ there exists a unique successor $A(x)$.*

Axiom 5 (Induction). If $1 \in S$ and $A(x) \in S$ for every $x \in S$, then $S = \mathbb{N}$.

Attention:

To prove $X(n)$ holds for every $n \in \mathbb{N}$: check the base case $n = 1$, then prove $X(n) \implies X(n+1)$.

Note:

The Hindu-Arabic decimal positional system we use today was developed in ***India****, later replacing Roman numerals in Europe.*

Meaning:

$S = \{y \in \mathbb{N} \mid x + y \text{ is well-defined}\}$ *collects every* y *for which the claim holds.*

1.1 Definition of addition

Theorem 1.1 (Existence and uniqueness of addition)**.** *For every* $x, y \in \mathbb{N}$ *there exists a unique* $x + y$ *satisfying:*

$$x + 1 = A(x), \qquad x + A(y) = A(x + y).$$

Proof. Fix x, let $S = \{y \mid x + y \text{ unique}\}$. Base: $y = 1$ gives $A(x)$, so $1 \in S$. Step: if $y \in S$ then $A(x + y) = x + A(y)$, so $A(y) \in S$. By induction, $S = \mathbb{N}$. □

Definition 1.1 (Sum)**.** The number $x + y$ above is called the **sum** of x and y.

Theorem 1.2.

$$1 + 3 = 4$$

Proof.

$$1 + 3 = 1 + A(2) = A(1 + 2) = A(A(2)) = A(3) = 4.$$

□

Theorem 1.3. *Commutativity of addition:*

$$x + y = y + x.$$

*This axiom lets us construct all other natural numbers: $A(1) = 2$, $A(2) = 3$, and so on.

Proof. Follows by double induction, first fixing $x = 1$, then applying the induction hypothesis on x. $\square$